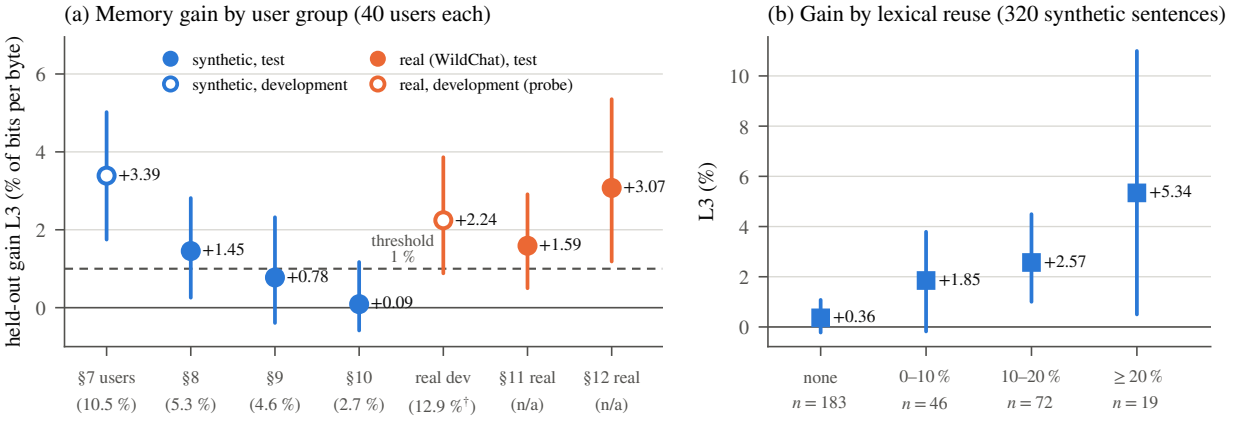
(a) Memory gain by user group (40 users each)
synthetic, test
synthetic, development
real (WildChat), test
real, development (probe)
held-out gain L3 (% of bits per byte)
6
4
2
0
+3.39
+1.45
+0.78
+0.09
threshold
1 %
+2.24
+1.59
+3.07
§7 users
(10.5 %)
§8
(5.3 %)
§9
(4.6 %)
§10
(2.7 %)
real dev
(12.9 %†)
§11 real
(n/a)
§12 real
(n/a)
(b) Gain by lexical reuse (320 synthetic sentences)
L3 (%)
10
8
6
4
2
0
+0.36
+1.85
+2.57
+5.34
none
n = 183
0–10 %
n = 46
10–20 %
n = 72
≥ 20 %
n = 19
share of a held-out sentence's bigrams already in the user's messages